

READMISSION RISK EXPLORER

Actionable intelligence at the exact moment of discharge.

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3.8 Million

US 30-day readmissions per year.

14.67%

National readmission rate (resulting in $2.5B in CMS penalties).

$52 Billion

Total annual system cost (AHRQ 2024).

A diabetic patient perfectly discharged today is back in the ER 30 days later. No warning. No intervention.

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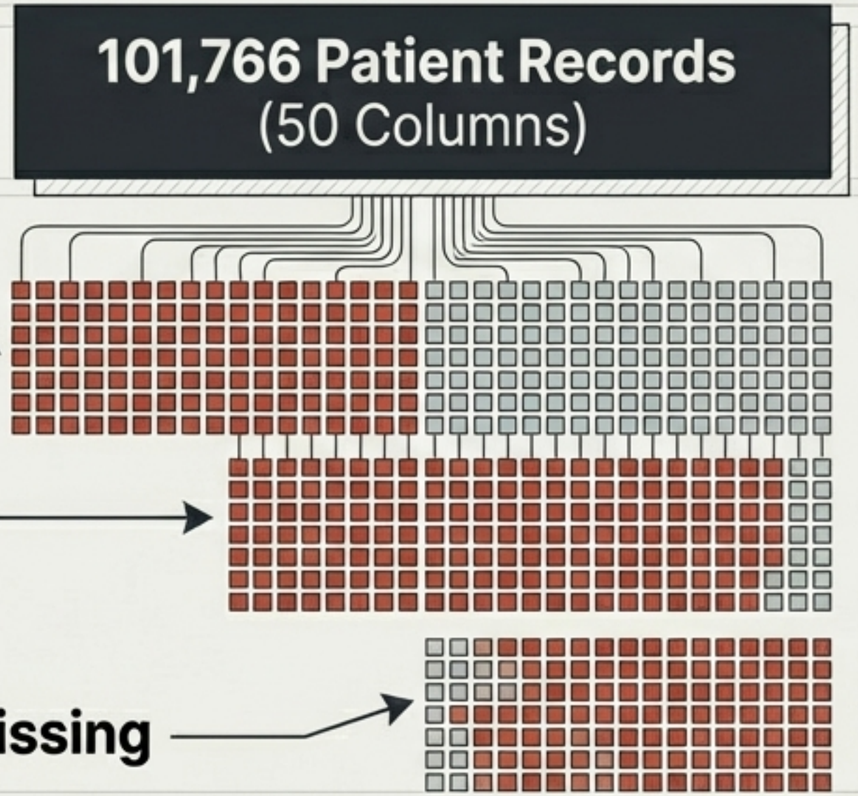
The Intelligence Gap

The Status Quo	Our Direction
✗ LACE scores used in complete isolation.	✓ Holistic, clinically grounded synthesis.
✗ Black-box ML models (zero explainability for doctors).	✓ Fully AI-explainable risk drivers.
✗ Paywalled AI APIs.	✓ Open-weight & On-premise (Strict HIPAA compliance).

The clinical failure isn't a lack of medical knowledge—it is a lack of timely synthesis.

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The Reality of EHR Data



Weight: 96.9% missing
(The most clinically relevant feature for diabetic prediction)

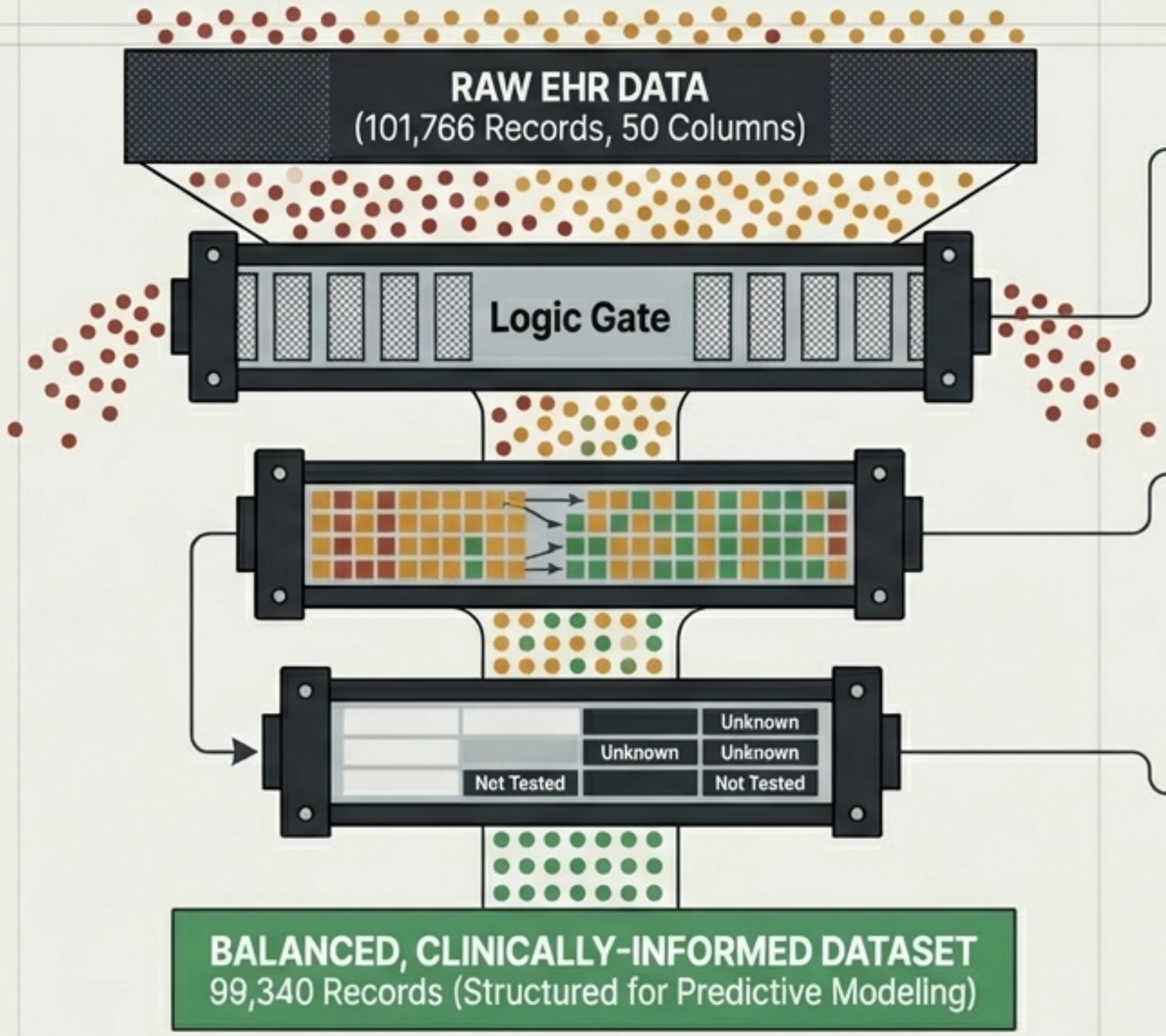
A1C Result: 83.1% missing
(The single most important lab marker for management)

Medical Specialty: 49.1% missing

This is not a perfectly curated academic dataset.
This behaves exactly like real, messy hospital EHR data.

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Clinical Thinking, Not Just Cleaning



Gate 1: Mortality Filter: Removed codes 11, 13, 14, 19, 20, 21 (Hospice/Expired)

Impact: 2,426 records dropped. **Rationale:** A deceased patient cannot be readmitted.

Gate 2: Target Reclassification: Merged <30 and >30 day readmissions.

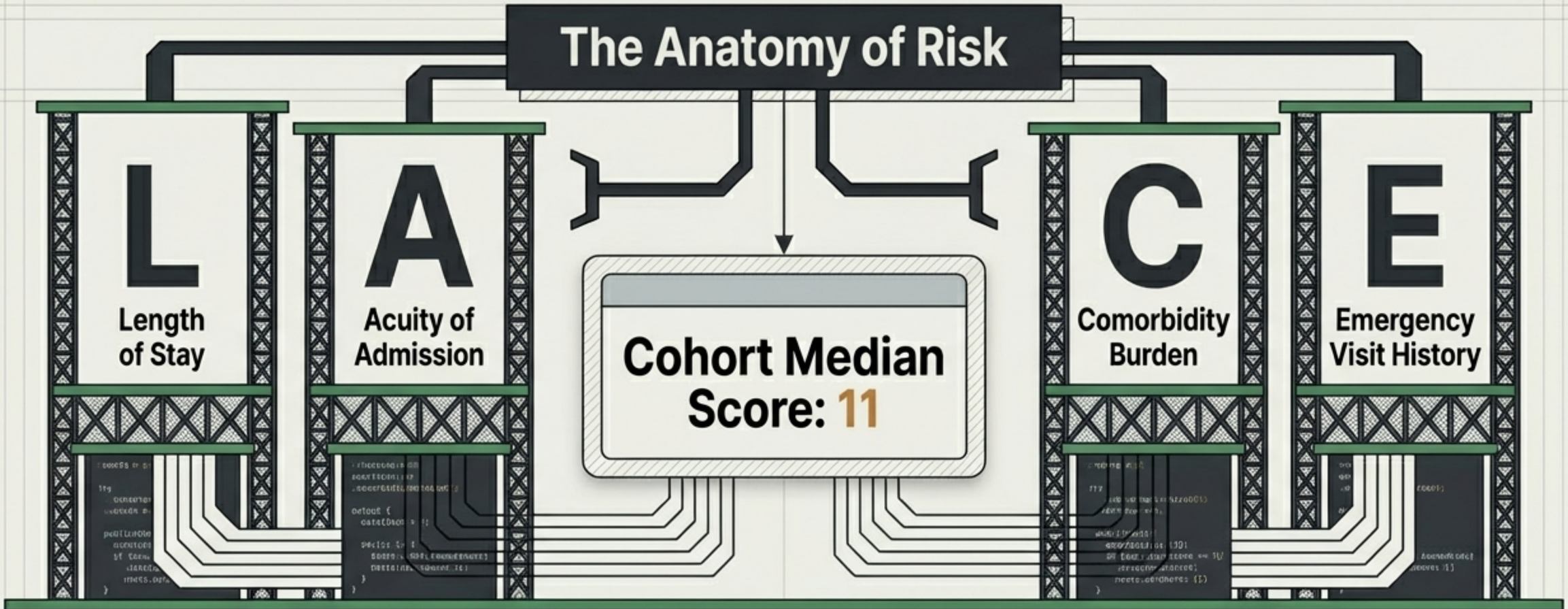
Impact: Created a 47–53% class split. **Rationale:** Any return is a missed intervention opportunity. No artificial oversampling needed.

Gate 3: Absence as Signal: Mapped missing Race, Specialty, and A1C to "Unknown"/"Not Tested".

Rationale: The absence of a test is itself a critical clinical signal.

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Feature Spotlight: Engineering LACE



Computed entirely from scratch. A median of 11 sits exactly at the threshold where real hospital intervention protocols activate.

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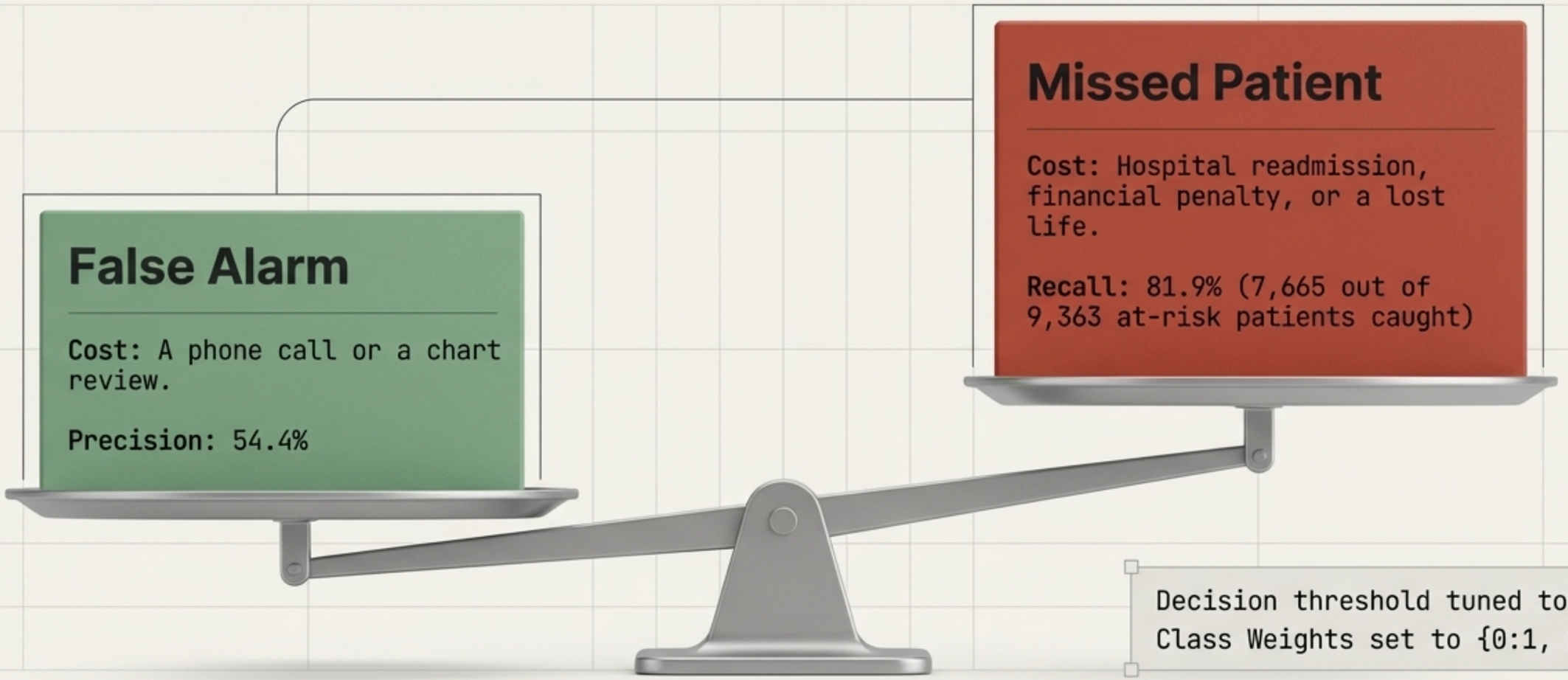
Model Selection Matrix

	COMPUTE COST	EXPLAINABILITY	PERFORMANCE (AUC)
XGBoost / Neural Networks	High Compute Cost	Black-box (No Explainability)	0.55 AUC
Random Forest	Medium Compute	Partial Explainability	0.60 AUC
✓ **Logistic Regression (Selected)**	**Low Compute**	**100% Explainable to Clinicians**	**0.62 AUC**

Choosing the simplest model wasn't a hackathon shortcut. On structured clinical data with significant missingness, research proves complex models degrade.

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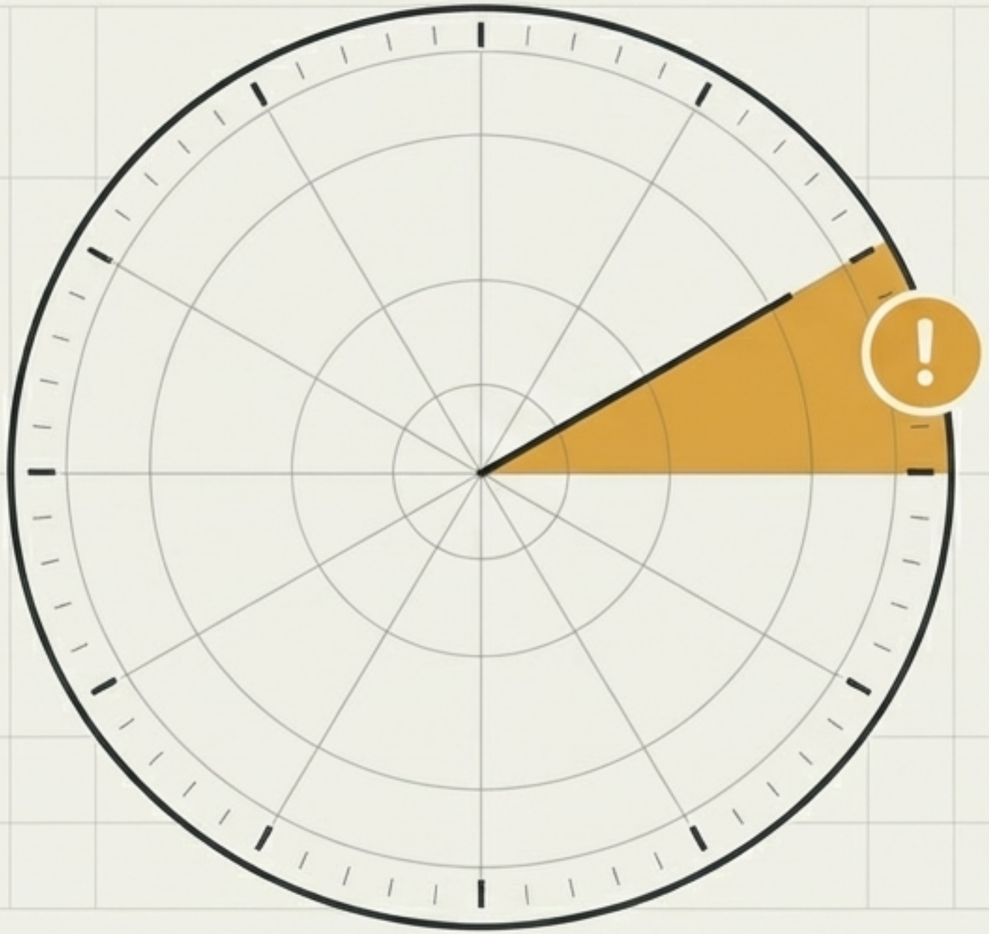
The Cost Equation



Missing a high-risk patient is twice as costly as a false alarm. We optimized for the right side of the equation.

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The 10-Minute Window



The Scenario

A doctor has 12 patients on her floor. She has exactly 10 minutes before discharge.

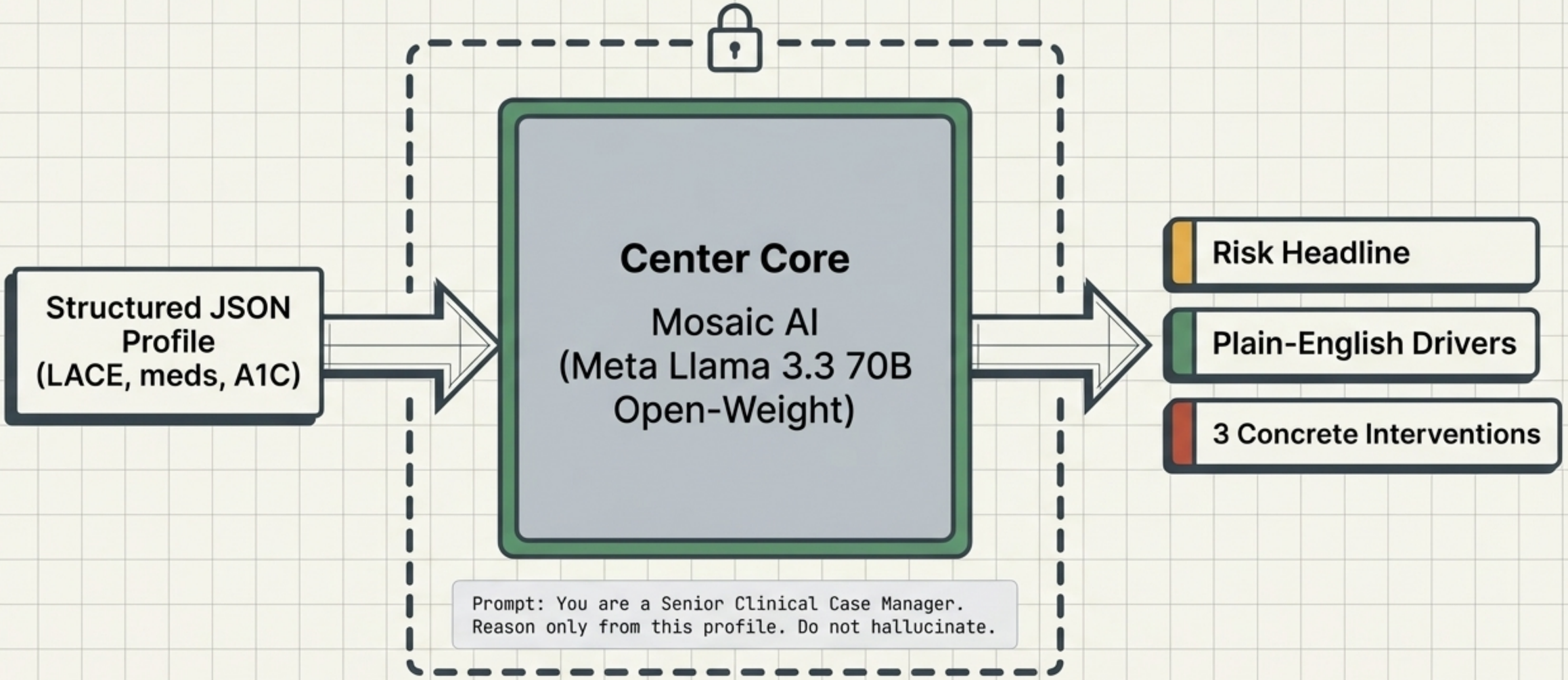
The Problem

The ML model flags a patient:
74% Risk Probability.
LACE Score **14**.
Insulin changed mid-stay.

She already knows her medicine. She doesn’t have time to dig through 50 columns of raw EHR data. She needs immediate, actionable synthesis.

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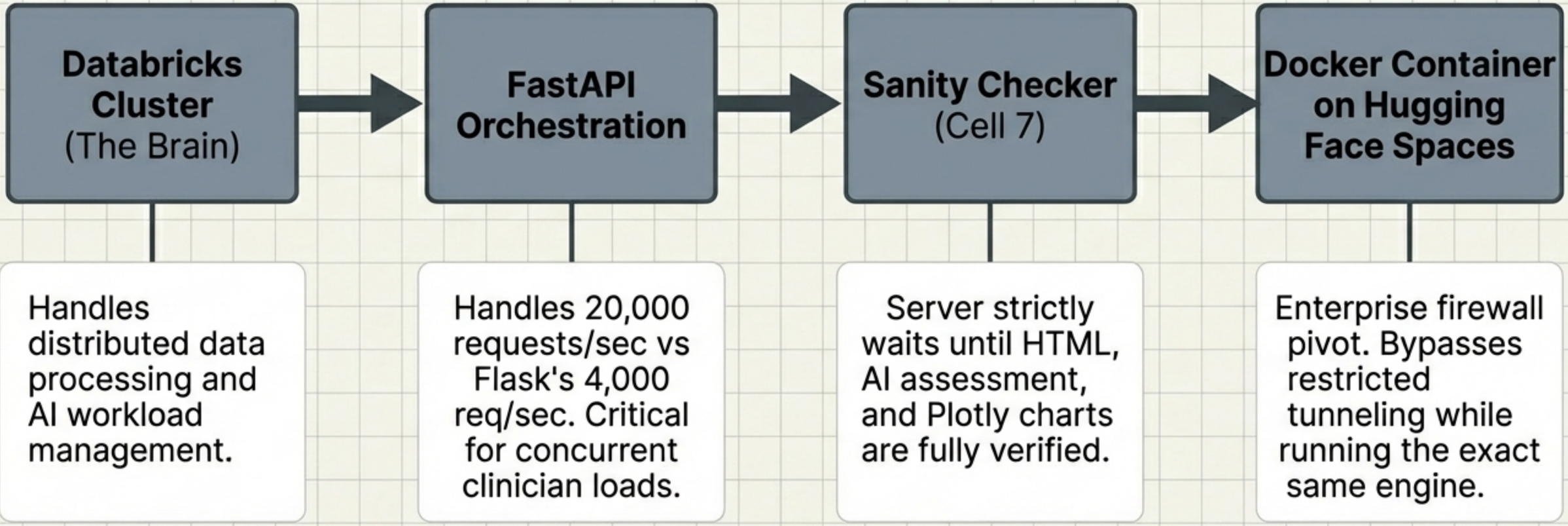
The Agentic AI Layer



Open-weight models run entirely on-premise. Patient data never leaves the building. This is a HIPAA compliance advantage, not a compromise.

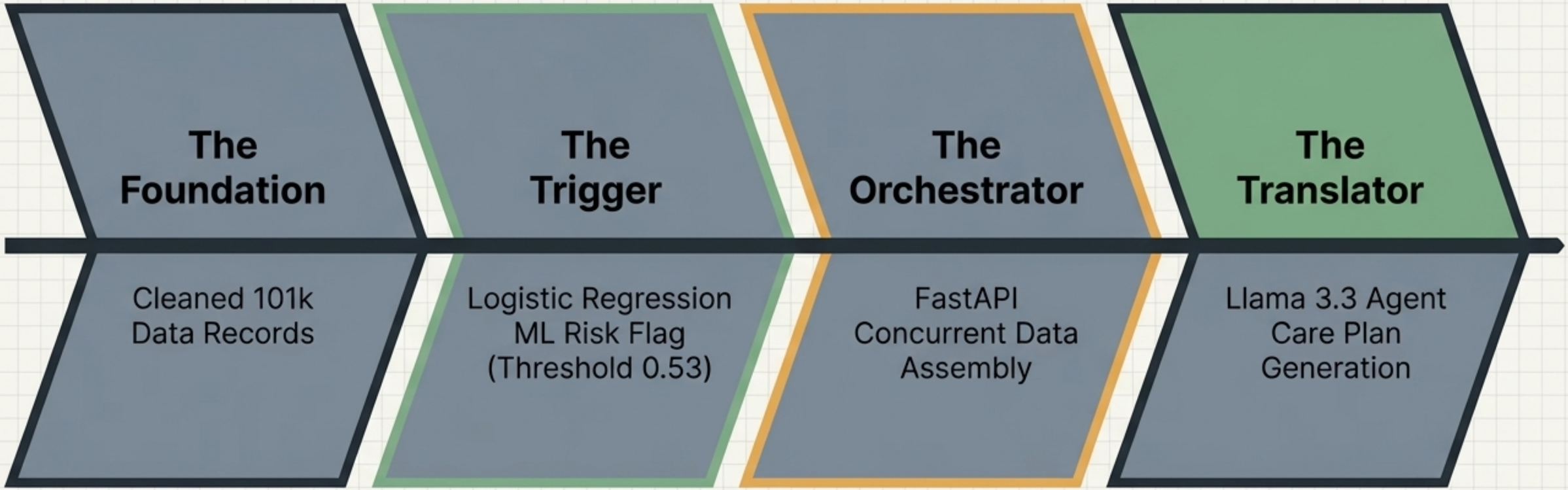
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Production-Grade Deployment Architecture



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Synthesis: The Intervention Engine



We didn’t build isolated scripts. We forged an end-to-end, functional clinical ecosystem.

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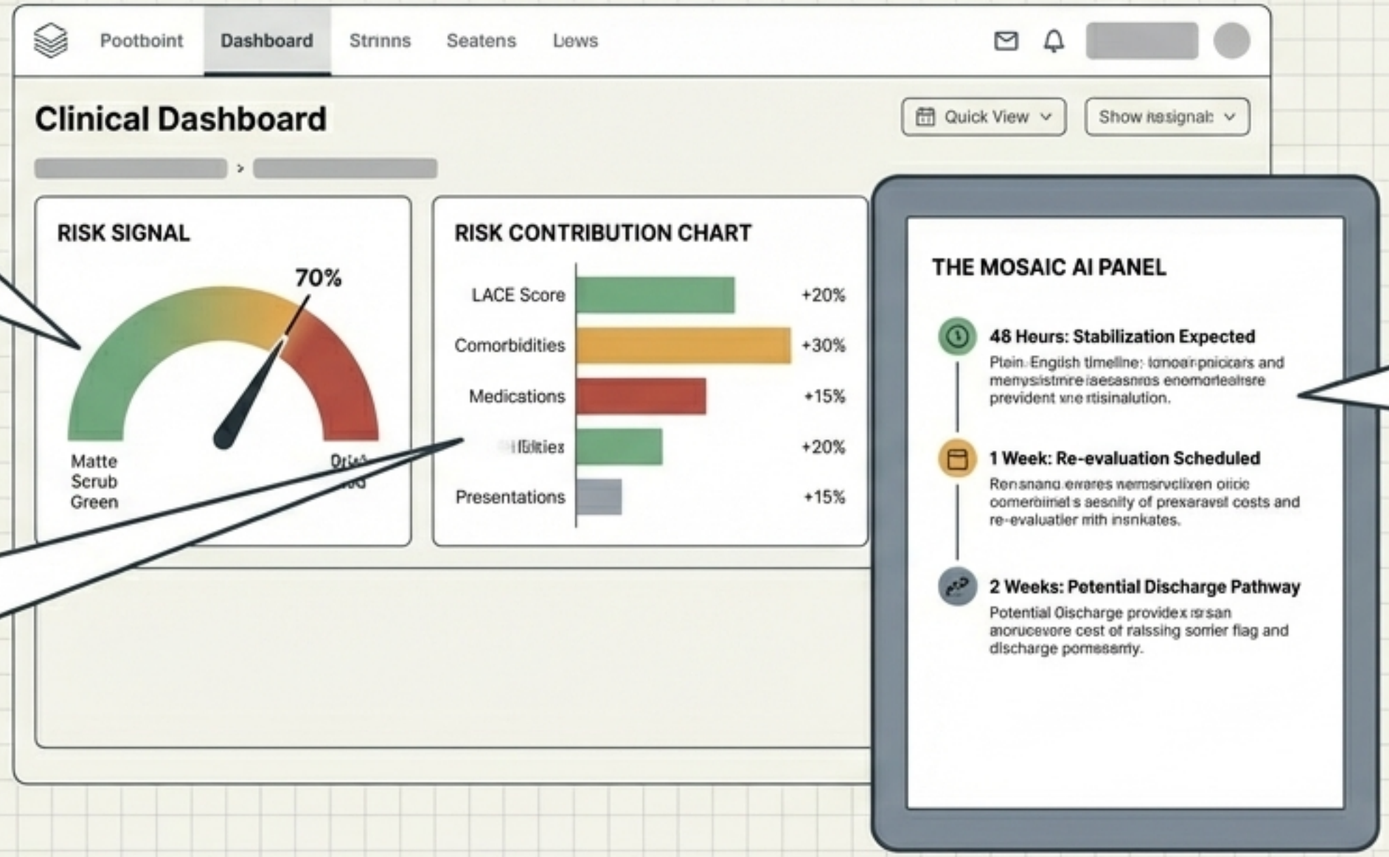
UI / The Dashboard Experience

The Risk Signal Gauge

Green to Critical Red dial, threshold at 70%

The Risk Contribution Chart

Horizontal impact bars & LACE Breakdown



The Mosaic AI Panel

Plain-English timeline:
48 hours, 1 week, 2 weeks

Transparent. Nothing hidden. A comprehensive care plan, not just a raw prediction number.

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The Concrete Next Steps



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“We didn’t just build a hackathon project. We built the first version of something a a hospital could actually use today.”



Scan to access the live deployment.

The architecture is real. The clinical logic is sound. Test it yourself right now.

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